Assignment 4: Neural Network Verification with alpha,beta-CROWN

Reliable and Trustworthy Artificial Intelligence | June 2026

1. Exploration Report: alpha,beta-CROWN Models and Configurations

The alpha,beta-CROWN repository provides 37 pre-trained models in complete_verifier/models/ and 263 YAML configuration files in complete_verifier/exp_configs/. Models span two datasets (MNIST, CIFAR-10) and multiple architectures: small fully-connected networks (toy/mnist_2_20.pth), CNNs from the ERAN benchmark (mnist_6_100, mnist_9_200, cifar_conv_small/big), adversarially trained CNNs from SDP (mnist_cnn_a_adv, cifar_cnn_a/b_adv), CIFAR-10 ResNets (resnet2b, resnet4b), non-ReLU models (sigmoid activations), and OVAL benchmark networks (cifar_base/deep/wide). All built-in weights use PyTorch format (.pth/.pt/.model); ONNX models are loaded at runtime via model.onnx_path in the YAML, not stored in this directory.

The 263 YAML configs are organised by verification method: beta_crown (27 files, core algorithm), bab_attack (6, combined verification + adversarial search), GCP-CROWN (13, Gurobi cut-plane), BICCOS (128, cut-enhanced for large models), and VNN-COMP competition sets (2021-2025). Each YAML has six top-level sections: model (path/name), data (dataset, mean/std, start/end indices), specification (norm, epsilon, or vnnlib_path for custom properties), attack (PGD pre-check), solver (batch_size, alpha/beta-crown iterations), and bab (timeout, branching). Unknown keys raise a ValueError at startup, making misconfiguration easy to detect.

Comparison with Marabou: Marabou requires constructing an explicit per-sample Python query with individual neuron bound variables, supports only ONNX input, and has no built-in dataset loader or PGD attacker. alpha,beta-CROWN uses a single YAML for the entire test set, natively loads MNIST/CIFAR, integrates a PGD attack as a fast pre-filter, and supports both PyTorch and ONNX models, ReLU and non-ReLU activations, and GPU acceleration.

2. Model, Dataset, and Verification Property

A fully-connected MLP (784->64->32->10, ReLU, no softmax) was trained on MNIST for 5 epochs with Adam (lr=1e-3), reaching 96.31% test accuracy. The model includes an explicit Flatten layer and was exported to ONNX (opset 14) with input shape (1,1,28,28) so alpha,beta-CROWN's built-in MNIST loader works directly. The same architecture was used in Assignment 3 (Marabou) to enable a direct performance comparison. Verification property: L-inf robustness with epsilon=0.01 on the first 100 MNIST test samples (indices 0-99, no normalisation).

3. Verification Results and Interpretation

alpha,beta-CROWN was run with kfsb branching, beta-CROWN 20 iterations, 60 s timeout, on CPU (no GPU). Table 1 summarises the 100-sample results.

Table 1. alpha,beta-CROWN results (100 MNIST samples, epsilon=0.01, L-inf)

Outcome	Count	%	Avg time (s)
safe-incomplete (CROWN bound sufficient)	97	97%	0.281
safe (BaB complete proof)	1	1%	0.567
unsafe-pgd (adversarial example found)	2	2%	0.008
timeout	0	0%	--
Total	100	100%	0.282

97 samples were verified by the CROWN relaxation alone in a single pass (safe-incomplete). One needed full BaB (sample 92). Two samples (indices 8 and 38) had adversarial examples found by PGD in under 13 ms each. For sample 38, the adversarial input is a digit image with up to 0.01 pixel perturbation that flips the prediction to class 3 (logits: [3.65, 2.59, 1.73, -0.52, ...]), indicating the model's decision boundary is unusually close to this test point. The 98% verified accuracy is high for an undefended MLP at epsilon=0.01.

4. Comparison: alpha,beta-CROWN vs Marabou

Table 2 compares both tools on the same task (same architecture, eps=0.01, L-inf, 100 MNIST samples). Note: the models are from independent training runs (same architecture, different random seeds), so slight differences in verified accuracy are expected.

Table 2. Tool comparison (same architecture, eps=0.01, 100 samples)

Metric	alpha,beta-CROWN	Marabou (A3)
Verified (safe)	98 / 100 (98%)	72 / 100 (72%)
Falsified (unsafe)	2 / 100 (2%)	28 / 100 (28%)
Timeout	0	0
Avg time / sample (s)	0.282	8.3
Speedup	~29x	1x
Model format	ONNX or PyTorch	ONNX only
Spec. format	YAML (norm, epsilon)	Python API per sample
Verification method	Bound prop + BaB	LP / SMT (ReLU split)

Speed: alpha,beta-CROWN is ~29x faster. CROWN resolves most samples in a single bound-propagation pass (~0.28 s), while Marabou's LP/SMT formulation requires iterative constraint solving for every ReLU split. Accuracy difference: the 26-point gap (98% vs 72%) is partly explained by different training runs and partly by alpha,beta-CROWN's PGD pre-check, which quickly falsifies borderline samples before invoking BaB. Ease of use: a single YAML controls the full pipeline in alpha,beta-CROWN; Marabou requires manual per-sample Python query construction.

5. Strengths and Limitations of alpha,beta-CROWN

Strengths

+ Speed via bound propagation:

CROWN linear relaxation proves most samples in a single pass, eliminating expensive BaB for the majority of instances.

+ Scalabi

Won multiple VNN-COMP iterations on networks far larger than this MLP. Handles CNNs and ResNets that would be intractable for Marabou.

+ Integra

Finds counterexamples in milliseconds before invoking formal proof, making falsification extremely fast.

+ Flexibl

One file controls model, dataset, epsilon, solver, and branching -- no per-sample code required.

Limitations

- Incomplete certificate:

safe-incomplete means the property is proven via a relaxation, not a rigorous BaB proof. For large epsilon or deep networks, BaB may time out.

- GPU de

Designed for CUDA. On CPU it is still fast for small models, but large networks benefit greatly from GPU availability.

- Comple

Requires conda + PyTorch + auto_LiRPA + onnx2pytorch. On Windows with non-ASCII paths, encoding workarounds in the test script were necessary. The YAML key reduceop (not reduce_op) and device:cpu for CPU-only runs were undocumented gotchas discovered during this assignment.